

Lecture Note 9: Uniformly Most Powerful Tests and the Neyman-Pearson Lemma

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1 Uniformly Most Powerful Test

Let $\Theta = \Theta_0 \cup \Theta_1$ be a parameter space. Consider a parametric family $\{f(x|\theta), \theta \in \Theta\}$. Suppose we want to test the null hypothesis, H_0 , that $\theta \in \Theta_0$ against the alternative, H_a , that $\theta \in \Theta_1$. Let C be some critical set. Then the probability that the null hypothesis is rejected is given by $\beta(\theta) = P_\theta\{X \notin C\}$. Recall that the test based on C is of level α if $\alpha \geq \sup_{\theta \in \Theta_0} \beta(\theta)$. The restriction of $\beta(\cdot)$ on Θ_1 is known as the power of the test. Let C' be another critical set. Denote the power of the test based on C' by $\beta'(\theta)$. Suppose that both tests are of level α . Then the test based on C is more powerful than the test based on C' if $\beta(\theta) \geq \beta'(\theta)$ for all $\theta \in \Theta_1$. Any test which is more powerful than any other test in some class $\mathcal{G}$ will be called the uniformly most powerful test in the class $\mathcal{G}$ (UMP test).

As follows from the theorem below, the UMP test exists if both the null and the alternative are simple.

Theorem 1 (Neyman-Pearson Lemma). *Let $f(x|\theta)$ with $\Theta = \{\theta_0, \theta_1\}$ be some parametric family. Suppose we want to test the null hypothesis, H_0 , that $\theta = \theta_0$ against the alternative hypothesis, H_a , that $\theta = \theta_1$. Assume that some critical set C satisfies (1) $x \in C$ if $kf(x|\theta_0) > f(x|\theta_1)$ and (2) $x \notin C$ if $kf(x|\theta_0) < f(x|\theta_1)$ where $k \geq 0$ is chosen so that $\alpha = P_{\theta_0}(X \notin C)$. Then the test based on C is the UMP test among all tests of level α . In addition, any UMP test of level α satisfies (1) and (2).*

Proof. Denote $\phi(x) = I\{x \notin C\}$, i.e. $\phi(x) = 1$ if $x \notin C$ and 0 otherwise. Thus, $\phi(x)$ denotes the probability that the test based on C rejects the null hypothesis upon observing data value x . Consider any other test of level α . Let $\tilde{\phi}(x)$ denote the probability that this test rejects the null hypothesis upon observing data value x . Since this test is of level α ,

$$\tilde{\beta}(\theta_0) = \int \tilde{\phi}(x)f(x|\theta_0)dx \leq \alpha,$$

where $\tilde{\beta}(\theta)$ denotes the probability that this test rejects the null hypothesis when the true parameter value is θ .

Note that

$$(\phi(x) - \tilde{\phi}(x))(f(x|\theta_1) - kf(x|\theta_0)) \geq 0$$

for any x . Indeed, if $f(x|\theta_1) - kf(x|\theta_0) > 0$, then $\phi(x) = 1$ and $\phi(x) - \tilde{\phi}(x) \geq 0$. If $f(x|\theta_1) - kf(x|\theta_0) < 0$, then $\phi(x) = 0$ and $\phi(x) - \tilde{\phi}(x) \leq 0$. If $f(x|\theta_1) - kf(x|\theta_0) = 0$, the product is zero. So,

$$0 \leq \int (\phi(x) - \tilde{\phi}(x))(f(x|\theta_1) - kf(x|\theta_0))dx = \beta(\theta_1) - \tilde{\beta}(\theta_1) - k(\beta(\theta_0) - \tilde{\beta}(\theta_0)),$$

where $\beta(\theta)$ denotes the probability that the test based on C rejects the null hypothesis when the true parameter value is θ . Therefore,

$$\beta(\theta_1) - \tilde{\beta}(\theta_1) \geq k(\beta(\theta_0) - \tilde{\beta}(\theta_0)) \geq k(\alpha - \alpha) = 0,$$

since $\tilde{\beta}(\theta_0) \leq \alpha$ and $\beta(\theta_0) = \alpha$. So the test based on C is more powerful than any other test of level α . So it is the UMP test, which proves the first statement of the theorem.

If $\tilde{\phi}(\cdot)$ is also a UMP test among all tests of level α , then $\tilde{\beta}(\theta_1) = \beta(\theta_1)$. So, $k(\alpha - \tilde{\beta}(\theta_0)) \leq 0$. Therefore $\tilde{\beta}(\theta_0) \geq \alpha$. On the other hand, $\tilde{\beta}(\theta_0) \leq \alpha$, since this test is of level α . We conclude that $\tilde{\beta}(\theta_0) = \alpha$. It follows that

$$\int (\phi(x) - \tilde{\phi}(x))(f(x|\theta_1) - kf(x|\theta_0))dx = 0.$$

Since the integrand is nonnegative for all x ,

$$(\phi(x) - \tilde{\phi}(x))(f(x|\theta_1) - kf(x|\theta_0)) = 0.$$

Thus, $\tilde{\phi}(x) = \phi(x)$ whenever $f(x|\theta_1) - kf(x|\theta_0) \neq 0$. So, $\tilde{\phi}(\cdot)$ also satisfies conditions (1) and (2). $\square$

Recall from the proof of the factorization theorem in Lecture Note 5 that if $T(X)$ is a sufficient statistic, then $f(x|\theta) = g(T(x)|\theta)h(x)$ where $g(\cdot)$ denotes the pdf of $T(X)$. So, in terms of the pdf of sufficient statistics, the critical set C of the UMP test satisfies (1) $x \in C$ if $kg(T(x)|\theta_0) > g(T(x)|\theta_1)$ and (2) $x \notin C$ if $kg(T(x)|\theta_0) < g(T(x)|\theta_1)$.

Example Let $X_1, \dots, X_n$ be a random sample from the $N(\mu, \sigma^2)$ distribution with known σ^2 . Suppose we want to test the null hypothesis, H_0 , that $\mu = \theta_0$ against the alternative hypothesis, H_a , that $\mu = \theta_1$. Without loss of generality we can assume that $\theta_0 > \theta_1$. We have already seen that the sufficient statistic in this example is given by $\bar{X}_n = \sum_{i=1}^n X_i/n$. We know that $\bar{X}_n \sim N(\mu, \sigma^2/n)$. So,

$$g(t|\theta) = C \exp\{-(n/(2\sigma^2))(t - \theta)^2\}.$$

From the Neyman-Pearson lemma, the UMP test among all tests of level α accepts the null hypothesis if and only if

$$kg(\bar{X}_n|\theta_0) > g(\bar{X}_n|\theta_1)$$

or, equivalently,

$$k \exp\{(n/(\sigma^2))\bar{X}_n(\theta_0 - \theta_1) - (n/(2\sigma^2))(\theta_0^2 - \theta_1^2)\} > 1.$$

Since $\theta_0 > \theta_1$, the test accepts the null hypothesis if and only if $\bar{X}_n > \tilde{k}$ where $\tilde{k}$ is such that $P_{\theta_0}(\bar{X}_n \leq \tilde{k}) = \alpha$. So, $\tilde{k} = \theta_0 + \sigma z_\alpha / \sqrt{n}$ where z_α denotes the α -quantile of the standard normal distribution.

Remark 1. The form of the optimal test can be derived directly from the optimization problem it solves. Write a (possibly randomized) test as a function $\phi(x) \in [0, 1]$, the probability of rejecting when the data equal x . The Neyman-Pearson problem is

$$\max_{\phi} \mathbb{E}_{\theta_1}[\phi(X)] \quad \text{subject to} \quad \mathbb{E}_{\theta_0}[\phi(X)] \leq \alpha.$$

For a multiplier $k \geq 0$, the Lagrangian is

$$\mathbb{E}_{\theta_1}[\phi(X)] - k(\mathbb{E}_{\theta_0}[\phi(X)] - \alpha) = \int \phi(x)(f(x; \theta_1) - kf(x; \theta_0)) dx + k\alpha,$$

which is maximized *pointwise*: set $\phi(x) = 1$ wherever $f(x; \theta_1) - kf(x; \theta_0) > 0$, i.e., wherever the likelihood ratio exceeds k , and $\phi(x) = 0$ wherever it falls below k ; the value of ϕ on the set $\{x: f(x; \theta_1) = kf(x; \theta_0)\}$ is where randomization may be needed, and k is chosen so that the size constraint binds. The cutoff form of the optimal test is therefore not a guess to be verified but the first-order condition of the problem, and the multiplier k has the usual interpretation as a shadow price: it measures the additional power that one more unit of size would buy.

2 UMP tests with composite hypotheses

The idea of the Neyman-Pearson lemma is to consider an optimization problem $\int \phi(x)f(x|\theta_1)dx \rightarrow \max$ subject to $\int \phi(x)f(x|\theta_0)dx \leq \alpha$ and $0 \leq \phi \leq 1$. The solution to this problem gives the UMP test of level α . This problem looks like maximization of utility given some budget constraint. We want to choose the most valuable items with the lowest price. In some special cases we can extend this idea to the case of composite hypotheses. Suppose we want to test the null hypothesis, H_0 , that $\theta \leq \theta_0$ against the alternative hypothesis, H_a , that $\theta > \theta_0$. Then the UMP test exists if $f(x|\theta)$ satisfies the monotone likelihood ratio property.

Definition 1. A family $f(x|\theta)$ with $\theta \in \mathbb{R}$ satisfies the monotone likelihood ratio if there exists some function $T(x)$ such that for any $\theta < \theta'$, $P_{\theta'}(x)/P_{\theta}(x)$ depends on x only through $T(x)$ and, moreover, $P_{\theta'}(x)/P_{\theta}(x)$ is a nondecreasing function of $T(x)$.

Theorem 2. Let $f(x|\theta)$ with $\theta \in \mathbb{R}$ be some parametric family that satisfies the monotone likelihood ratio with function $T(x)$. Suppose we want to test the null hypothesis, H_0 , that $\theta \geq \theta_0$ against the alternative hypothesis, H_a , that $\theta < \theta_0$. Then a UMP test of level α exists. It is given by $\phi(X) = 1$ if $T(X) < C$, $\phi(X) = \gamma$ if $T(X) = C$, and 0 otherwise, for some constants C and γ such that $\mathbb{E}_{\theta_0}[\phi(X)] = \alpha$. In addition, the power of this test $\beta(\theta) = \mathbb{E}_{\theta}[\phi(X)]$ for $\theta > \theta_0$ is strictly decreasing in θ .

Proof. Choose a simple alternative $\theta_1 < \theta_0$. By the Neyman-Pearson lemma, the UMP test of θ_0 against θ_1 accepts the null hypothesis if $f(X|\theta_0)/f(X|\theta_1) > k$ and rejects it if $f(X|\theta_0)/f(X|\theta_1) < k$. By the monotone likelihood ratio, this test accepts the null hypothesis if $T(X) > C$ and rejects it if $T(X) < C$. When $T(X) = C$, the test rejects with probability, say, γ . The constants C and γ should be chosen such that $\mathbb{E}_{\theta_0}[\phi(X)] = \alpha$. Now, note that the same test will be UMP of level α

for any other alternative $\theta_2 < \theta_0$ as well. So, this test is UMP of level α for the null hypothesis $\theta = \theta_0$ against the alternative $\theta < \theta_0$. Thus, to show that the same test is UMP of level α for the null $\theta \geq \theta_0$ against the alternative $\theta < \theta_0$, it will be enough to show that this test is of level α , i.e. $\sup_{\theta \geq \theta_0} \beta(\theta) \leq \alpha$.

Since there always exists a test of level α with power α (this test rejects the null hypothesis with probability α independently of the data), $\beta(\theta_0) \leq \beta(\theta_1)$. Since the test based on $T(x)$, C , and γ is also UMP (of some level) for the null hypothesis $\theta = \theta_1$ against the alternative $\theta = \theta_2$ for any $\theta_2 < \theta_1$, the same argument yields $\beta(\theta_1) \leq \beta(\theta_2)$ for any $\theta_1 \geq \theta_2$, which is the second statement of the theorem. The first statement of the theorem follows from $\sup_{\theta \geq \theta_0} \beta(\theta) = \beta(\theta_0) = \alpha$ since $\beta(\theta)$ is decreasing in θ . $\square$

In many cases a UMP test does not exist. Below is an example of such a situation.

Example Let $X_1, \dots, X_n$ be a random sample from an $N(\theta, \sigma^2)$ distribution with known σ^2 . Suppose we want to test the null hypothesis, H_0 , that $\theta = \theta_0$ against the alternative hypothesis, H_a , that $\theta \neq \theta_0$. Consider some $\theta_1 < \theta_0$. The only UMP test of level α of $\theta = \theta_0$ against $\theta = \theta_1$ rejects the null hypothesis if and only if $\bar{X}_n < \theta_0 + \sigma z_\alpha / \sqrt{n}$, as we have already seen. But this test has little power in our situation for any $\theta > \theta_0$. Indeed, $\beta(\theta) < \alpha$ for all $\theta > \theta_0$. So, this test cannot be UMP in this situation, since there always exists a test of level α with power α .

2.1 Unbiased tests

Since there are no UMP tests among all tests of level α in many situations, the question arises whether we can find UMP tests in some smaller, but still reasonably large, classes of tests. The definition below gives a property that reasonable tests should have.

Definition 2. Any test of the null hypothesis $\theta \in \Theta_0$ against the alternative $\theta \in \Theta_1$ is called unbiased if for some $\alpha \in [0, 1]$, $\beta(\theta) \leq \alpha$ for all $\theta \in \Theta_0$ and $\beta(\theta) \geq \alpha$ for all $\theta \in \Theta_1$.

Let $\Theta = \mathbb{R}$ be a parameter space. Suppose we want to test the null hypothesis, H_0 , that $\theta = \theta_0$ against the alternative hypothesis, H_a , that $\theta \neq \theta_0$. Consider a test which rejects the null hypothesis with probability $\phi(x)$ upon observed data value x . As before, denote $\beta(\theta) = \int \phi(x) f(x|\theta) dx$. If $\beta(\theta)$ is differentiable in θ , then for any unbiased test, we necessarily have $\beta'(\theta_0) = 0$. Indeed, if this condition does not hold, then there exists a point θ in a neighborhood of θ_0 such that $\beta(\theta) < \beta(\theta_0)$ by definition of derivative. In some situations, there exists a UMP test among all unbiased tests of level α even though there are no UMP tests among all tests of level α . In particular, for the Gaussian mean model considered above, imposing unbiasedness resolves the two-sided problem: the UMP test among all unbiased tests of level α rejects the null hypothesis if and only if $|\bar{X}_n - \theta_0|$ exceeds a critical value, i.e. the familiar two-sided test.

Remark 2. The Neyman-Pearson Lemma is also the fundamental result on *targeting* with limited resources. Suppose a regulator can audit at most a fraction α of firms, and wishes to catch as

many non-compliant firms as possible; compliant firms generate data X from $f(x; \theta_0)$ and non-compliant firms from $f(x; \theta_1)$. An audit rule is exactly a test $\phi(x) \in [0, 1]$, its size is the audit rate among compliant firms, and its power is the detection rate. The Lemma says: rank firms by the likelihood ratio and audit from the top down — no feasible rule detects more. In the language of classification, among all rules with false-positive rate at most α , thresholding the likelihood ratio maximizes the true-positive rate; varying α traces out the best attainable frontier of true-positive against false-positive rates (the “ROC curve” of machine learning). Modern prediction algorithms used to target audits, inspections, or interventions are, in effect, estimating a likelihood ratio and thresholding it at capacity — the 1933 lemma is the reason that recipe is the right one.

3 Likelihood Ratio Test

Suppose we want to test the null hypothesis, H_0 , that $\theta \in \Theta_0$ against the alternative hypothesis, H_a , that $\theta \in \Theta_1$. Denote $\Theta = \Theta_0 \cup \Theta_1$. Let $\mathcal{L}(\theta|x)$ denote the likelihood function. Then we have

Definition 3. A likelihood ratio test (LRT) statistic is

$$\lambda(x) = \frac{\sup_{\theta \in \Theta_0} \mathcal{L}(\theta|x)}{\sup_{\theta \in \Theta} \mathcal{L}(\theta|x)}.$$

By definition, $0 \leq \lambda(x) \leq 1$. Small values of the LRT statistic imply that there is a value θ in the alternative hypothesis Θ_1 which gives much greater likelihood than all values in the null hypothesis. So, likelihood ratio tests reject the null hypothesis if and only if $\lambda(x) \leq c$ for some c . As usual, the constant c is chosen according to the desired level of the test.

Let $\hat{\theta}_r = \arg \max_{\theta \in \Theta_0} \mathcal{L}(\theta|x)$ be the ML estimator of the restricted model. Let $\hat{\theta}_{ur} = \arg \max_{\theta \in \Theta} \mathcal{L}(\theta|x)$ be the ML estimator of the unrestricted model. Then an equivalent way to define the LRT statistic is to set

$$\lambda(x) = \frac{\mathcal{L}(\hat{\theta}_r|x)}{\mathcal{L}(\hat{\theta}_{ur}|x)}.$$

Example Let $X_1, \dots, X_n$ be a random sample from an $N(\theta, 1)$ distribution. Suppose we want to test the null hypothesis, H_0 , that $\theta = \theta_0$ against the alternative hypothesis, H_a , that $\theta \neq \theta_0$. Then $\hat{\theta}_r = \theta_0$ and $\hat{\theta}_{ur} = \hat{\theta}_{MLE} = \bar{X}_n$. So, the LRT statistic is

$$\begin{aligned} \lambda(x) &= \frac{\mathcal{L}(\theta_0|x)}{\mathcal{L}(\hat{\theta}_{MLE}|x)} \\ &= \frac{(2\pi)^{-n/2} \exp\{-(1/2) \sum_{i=1}^n (X_i - \theta_0)^2\}}{(2\pi)^{-n/2} \exp\{-(1/2) \sum_{i=1}^n (X_i - \bar{X}_n)^2\}} \\ &= \exp\{-(1/2) \sum_{i=1}^n [(X_i - \bar{X}_n + \bar{X}_n - \theta_0)^2 - (X_i - \bar{X}_n)^2]\} \\ &= \exp\{-(n/2)(\bar{X}_n - \theta_0)^2\} \end{aligned}$$

So, the LRT rejects the null hypothesis if and only if $|\bar{X}_n - \theta_0| > c$. Specifically, the LRT of level α rejects the null hypothesis if and only if $\bar{X}_n - \theta_0 > z_{1-\alpha/2}/\sqrt{n}$ or $\bar{X}_n - \theta_0 < z_{\alpha/2}/\sqrt{n}$ since under the null hypothesis, $\bar{X}_n \sim N(\theta_0, 1/n)$.